

PAUL SENTONGO

Data Scientist • Applied AI Researcher • MSc Data Science

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RESEARCH PROFILE

Applied AI researcher with a production engineering background in high-stakes environments. My work sits at the intersection of explainability, clinical decision support, and responsible deployment building ML systems for healthcare and financial services that are interpretable by design, not as an afterthought. Three years of industry experience in regulated AI (banking, Absa Uganda) and an independent portfolio of deployed medical AI systems ground my research interests in the practical failure modes that formal methods in trustworthy AI need to solve.

Explainable AI (XAI) · **Trustworthy AI** · **Continual Learning** · **Medical Image Analysis** · **Clinical Decision Support** · **Responsible AI in High-Stakes Systems**

EDUCATION

MSc Data Science and Analytics

2023 – Oct 2025

Uganda Christian University

Mukono, Uganda

CGPA: 4.16 / 5.0 (equivalent 8.32 / 10) **Distinction** Major: Machine Learning for Health Informatics & Computer Vision

- **Thesis:** “Machine Learning for Accurate Valuation of Imported Items in Uganda Customs” — published in the UCU Digital Repository: ucudir.ucu.ac.ug/items/a422584f-96fb-481e-90c5-324800fadc75
- Relevant modules: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Statistical Modelling, Big Data Analytics, Research Methods in Data Science

BSc Business Computing

2011 – Oct 2014

Uganda Christian University

Mukono, Uganda

CGPA: 3.53 / 5.0 **Second Class Honours** Major: Business Intelligence & Data Warehousing

THESIS AND PUBLICATIONS

Sentongo, P. (2025). *Machine Learning for Accurate Valuation of Imported Items in Uganda Customs*. MSc Thesis, Uganda Christian University.

Published in the UCU Digital Repository: ucudir.ucu.ac.ug/items/a422584f-96fb-481e-90c5-324800fadc75

RESEARCH PROJECTS — APPLIED AI SYSTEMS

Nakaseke NCD-AI: Community Hypertension Risk Screener

2026

End-to-end ML system for questionnaire-only hypertension screening • Real clinical data • N = 3,471 patients • github.com/sentongo-web/Diabetes-Detection-Complete-MLOPs

SHAP • **Scikit-learn** • **Optuna** • **Flask** • **Docker** • **MLflow** • **SMOTE**

- Built a hypertension risk model from 3,471 real patient records collected at Nakaseke Hospital,

Uganda, computing ground-truth labels from three averaged clinical blood pressure readings using the **WHO/JNC-8 definition** rather than self-reported data explicitly addressing recall bias and reporting bias in field-collected clinical data.

- Trained and evaluated six candidate models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM, KNN) under a rigorous three-way stratified split; **SMOTE applied strictly inside training folds** to prevent data leakage; Optuna Bayesian hyperparameter search (60 trials, TPE) for the top-two candidates.
- Achieved test **AUC 0.639** consistent with published literature for questionnaire-only hypertension screening in LMICs (0.60–0.72) deliberately excluding blood pressure readings as model inputs, since their presence would make the tool clinically useless as a screener.
- Designed a **SHAP-based explanation layer** grounded in game-theoretic guarantees (efficiency, symmetry, linearity) that produces personalised clinical justifications per prediction an implementation of XAI for non-expert clinical stakeholders in low-resource settings.
- Deployed as a Flask web application serving **calibrated risk probabilities** (not hard labels), with zero patient data storage and Docker containerisation for reproducible deployment.

2026

KidneyDL: Medical Image Classification with Full MLOps Pipeline

4-class kidney CT scan classifier • Transfer learning • Live deployment • Complete CI/CD • huggingface.co/spaces/Sentoz/kidney-ct-classifier

ResNet-50 • TensorFlow • DVC • MLflow • DagsHub • Docker • GitHub Actions • Hugging Face

- Trained a **4-class kidney CT scan classifier** (Normal / Cyst / Stone / Tumour) using ResNet-50 fine-tuned on a curated CT dataset extending beyond binary detection to reflect real diagnostic complexity that a clinical support tool must handle.
- Selected ResNet-50 deliberately for its residual skip connections and proven performance on fine-grained texture discrimination in medical imaging; architecture choice documented with clinical rationale **behavioural predictability treated as a safety property**, not just a performance metric.
- Implemented a **four-stage DVC-tracked pipeline** (data ingestion → base model preparation → training → evaluation) with explicit dependency declarations, ensuring only invalidated stages rerun and the full pipeline is reproducible from any state.
- All experiments tracked remotely via **MLflow on DagsHub**; models versioned in the MLflow Model Registry; **GitHub Actions CD pipeline** deploys to Hugging Face Spaces on every push to main — demonstrating the full responsible AI deployment chain: artifact versioning, containerised serving, automated deployment.
- Achieves **89.9% validation accuracy**; architecture selected for interpretability and predictability — properties that matter as safety requirements in medical AI, not only performance ones.

2026

Dental OPG Cavity Detection: Object Detection for Clinical Screening

YOLOv8 object detection on full-mouth panoramic radiographs • Live deployment • huggingface.co/spaces/Sentoz/dental-opg-cavity-detection

YOLOv8 • PyTorch • Object Detection • Gradio • Hugging Face Spaces

- Developed a **YOLOv8 object detection pipeline** for automated cavity detection in Orthopantomographic (OPG) dental panoramic radiographs a harder task than classification due to simultaneous multi-region detection requirements across variable-density radiograph anatomy.
- The system draws **confidence-scored bounding boxes** around suspicious cavity regions, producing immediately interpretable visual output; confidence scores surfaced rather than suppressed, giving the clinician calibrated uncertainty information to act on a concrete implementation of the **human-in-the-loop principle** that trustworthy AI frameworks require for medical decision support.
- Design choices reflect responsible deployment priorities: the tool explicitly positions itself as a screening aid, not a diagnosis; no patient images are stored; built for low-resource clinical settings

across sub-Saharan Africa where specialist radiology access is limited.

PROFESSIONAL EXPERIENCE

Data Scientist

April 2025 – Present

Kampala, Uganda

Absa Bank Uganda Limited

- Developed and validated predictive models for customer churn risk and fraud detection in a regulated financial environment, handling severe class imbalance using the same SMOTE-based techniques applied in the Nakaseke NCD-AI project. Grew the customer base by **18%** and reduced fraudulent transactions by **30%**.
- Implemented **SHAP-based explanation layers** for credit risk models to satisfy regulatory requirements; designed structured natural-language justifications for loan officers operationalising XAI for non-expert stakeholders under compliance constraint.
- Built **drift-monitoring pipelines** (KS-test, PSI) tracking feature distributions and output calibration across production models; flagged distributional shift caused by macroeconomic shocks and triggered structured retraining workflows practical continual learning under regulatory constraint.
- Conducted **demographic fairness audits** across credit scoring models, identifying disparate impact patterns and working with compliance teams to revise model thresholds and document limitations surfacing the gap between statistical fairness metrics and regulatory fairness obligations.
- Contributed to a **22% reduction in turnaround time** through exploratory and root-cause analysis on structured and unstructured datasets.

Operations Business and Performance Analyst

Aug 2024 – April 2025

Kampala, Uganda

Absa Bank Uganda Limited

- Designed and deployed automated data pipelines for KPI and management information reporting, reducing manual effort by **35%** and consolidating fragmented operational data sources into a single auditable system.
- Built executive dashboards tracking productivity and SLA compliance; data-driven interventions improved operational productivity by **30%**.
- Developed a real-time system downtime monitoring web application with automated alert triggering — reducing operations downtime by **20%**.
- Drove root-cause analysis on SLA breaches and queue-time patterns, enabling process redesigns that cut average processing time by **22%**.

Acting Branch Manager

Jan 2024 – Aug 2024

Kampala, Uganda

Absa Bank Uganda Limited

- Led branch to a clean audit with **zero high-risk findings**, demonstrating rigorous process governance and quality-assurance standards directly relevant to AI conformity assessment.
- Managed cross-functional team driving **18% sales growth** and **12% deposit growth**; standardised cash-operations controls reducing cash-variance incidents by **40%**.

TECHNICAL SKILLS

Deep learning frameworks: TensorFlow, Keras, PyTorch; CNNs, ResNet-50, VGG16, YOLOv8 fine-tuning; transfer learning

Explainability & fairness: SHAP (SHapley Additive Explanations), Grad-CAM, demographic bias

auditing, precision-recall optimisation

MLOps & engineering: DVC, MLflow, DagsHub, Docker, GitHub Actions CI/CD, Hugging Face Spaces, Gradio, FastAPI, Flask, REST API design

Applied ML methods: Classification, object detection, regression, SMOTE, class-imbalance handling, Optuna hyperparameter tuning, cross-validation, concept drift detection (KS-test, PSI)

Medical imaging: CT scan classification (4-class), OPG panoramic radiograph object detection, bounding-box localisation, image augmentation

Languages & data: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, Stata (.dta), YAML configuration management

Cloud & infrastructure: GCP, AWS, Docker Compose, VS Code, Google Colab, Power BI, Jupyter Notebooks

PROFESSIONAL AFFILIATIONS

- Ambassador, **World Ethical Data Foundation** representing responsible AI and data governance principles (worldethicaldata.org/projects/ambassadorship)

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- PCAP: Python Programming Fundamentals — The Python Institute
- Artificial Intelligence A-Z — Udemy (deep reinforcement learning, neural architectures)
- NLP with BERT, GPT and Transformers — Udemy (transformer architectures and fine-tuning)
- Python Bootcamp: Zero to Hero — Udemy
- Big Data and Hadoop: The Complete Guide — Udemy
- SQL for Business Intelligence and Reporting — Udemy
- Cloud Computing and AWS: Complete Bootcamp — Udemy

REFERENCES

Mr Ian Raymond Osolo

Academic Supervisor

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Mr Andrew Kisawuzi

Data Engineer

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